Maharashtra Education Society's

Institute of Management and Career Courses (IMCC), Pune

AUTONOMOUS

IT13: ADBMS

Chapter 2:SQL

Practical Assignment 1 Solution Template

Name: Mohit Dhumal

Roll No: 2601051

Practical 1. Implementation of DDL & DML commands in SQL with suitable examples

Create the following tables. (Primary Keys are underlined)

Employee (emp_id ,emp_name, City, bdate)

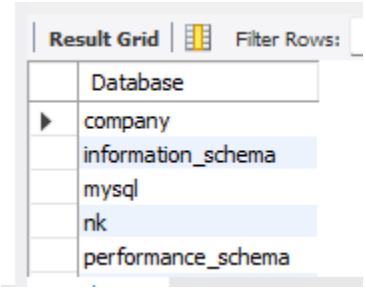
Investor (inv_no, inv_name, inv_date, inv_amt, emp_id)

Create a Primary key constraint on emp_id in the Employee table and inv_no in the Investor table.

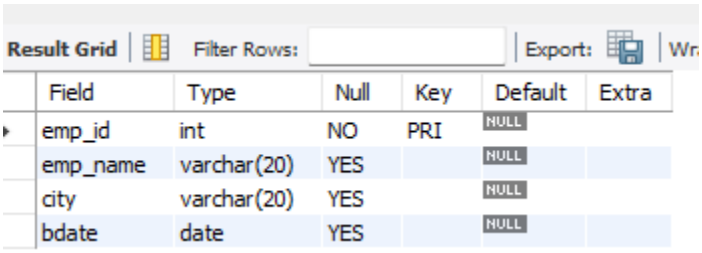
Write the SQL query and paste a screenshot of the output below the question.

1. Create a database and two tables in it.

```
CREATE DATABASE company;
USE company;
SHOW DATABASES;
```

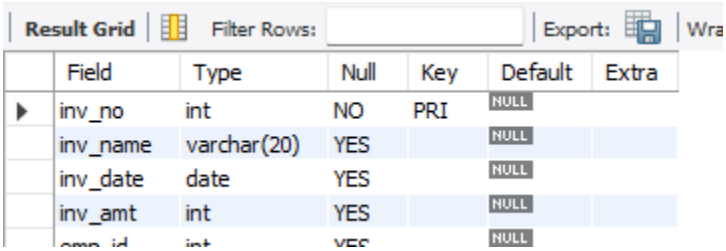



```
CREATE TABLE Employee(
                emp_id INT PRIMARY KEY,
    emp_name VARCHAR(20),
    city VARCHAR(20),
    bdate DATE
);
DESC Employee;
```



Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	
emp_name	varchar(20)	YES		NULL	
city	varchar(20)	YES		NULL	
bdate	date	YES		NULL	

```
CREATE TABLE Investor(
        inv_no INT PRIMARY KEY,
    inv_name VARCHAR(20),
    inv_date DATE,
    inv_amt INT,
    emp_id INT
);
DESC Investor;
```



Field	Type	Null	Key	Default	Extra
inv_no	int	NO	PRI	NULL	
inv_name	varchar(20)	YES		NULL	
inv_date	date	YES		NULL	
inv_amt	int	YES		NULL	

2. Insert at least 5 rows into the table.

```
INSERT INTO Employee(emp_id,emp_name,city,bdate) VALUES
(101,'Mohit','Pune','2005-10-12'),
(102,'Nikhil','Mumbai','2005-08-08'),
(103,'Parnika','Nashik','2005-07-30'),
(104,'Yash','Kolhapur','2005-01-26'),
(105,'Riya','Chennai','2005-08-15');
SELECT * FROM Employee;
```

Result Grid | Filter Rows:

emp_id	emp_name	city	bdate
101	Mohit	Pune	2005-10-12
102	Nikhil	Mumbai	2005-08-08
103	Parnika	Nashik	2005-07-30
104	Yash	Kolhapur	2005-01-26
105	Riya	Chennai	2005-08-15
NULL	NULL	NULL	NULL

INSERT INTO Investor(inv_no,inv_name,inv_date,inv_amt,emp_id) VALUES
(1,'Vikas','2026-08-24',1000,101),
(2,'Manoj','2025-10-05',55000,101),
(3,'Rahul','2024-02-09',78000,102),
(4,'Virat','2025-09-30',500,103),
(5,'Shardul','2026-04-21',34000,104);

SELECT * FROM Investor;

Result Grid | Filter Rows: | Edit:

inv_no	inv_name	inv_date	inv_amt	emp_id
1	Vikas	2026-08-24	1000	101
2	Manoj	2025-10-05	55000	101
3	Rahul	2024-02-09	78000	102
4	Virat	2025-09-30	500	103
5	Shardul	2026-04-21	34000	104
NULL	NULL	NULL	NULL	NULL

3. Display all the information in the Employee table.

SELECT * FROM Employee;

Result Grid | Filter Rows:

emp_id	emp_name	city	bdate
101	Mohit	Pune	2005-10-12
102	Nikhil	Mumbai	2005-08-08
103	Parnika	Nashik	2005-07-30
104	Yash	Kolhapur	2005-01-26
105	Riya	Chennai	2005-08-15
NULL	NULL	NULL	NULL

4. Add columns 'Age' and 'Salary' to the Employee table; apply check constraints for (Age<=50 and Salary>=10000).

ALTER TABLE Employee ADD COLUMN age INT;

ALTER TABLE Employee ADD COLUMN salary INT;

DESC Employee;

Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	
emp_name	varchar(20)	YES		NULL	
city	varchar(20)	YES		NULL	
bdate	date	YES		NULL	
age	int	YES		NULL	
salary	int	YES		NULL	

ALTER TABLE Employee ADD CONSTRAINT chk_age check(age<=50);

ALTER TABLE Employee ADD CONSTRAINT chk_salary check(salary>=10000);

79	14:45:06	ALTER TABLE Employee ADD CONSTRAINT chk_age check(age<=50)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.031 sec
80	14:45:08	ALTER TABLE Employee ADD CONSTRAINT chk_salary check(salary>=10000)	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.032 sec

5. Add a column 'Age' to the Investor table.

ALTER TABLE Investor ADD COLUMN age INT;

DESC Investor;

Result Grid | Filter Rows: | Export: | Wrap Cell Conten

Field	Type	Null	Key	Default	Extra
inv_no	int	NO	PRI	NULL	
inv_name	varchar(20)	YES		NULL	
inv_date	date	YES		NULL	
inv_amt	int	YES		NULL	
emp_id	int	YES		NULL	
age	int	YES		NULL	

6. Change the data type of inv_name from varchar to text in the Investor table

ALTER TABLE Investor MODIFY COLUMN inv_name TEXT;

DESC Investor;

Result Grid | Filter Rows: | Export:

Field	Type	Null	Key	Default	Extra
inv_no	int	NO	PRI	NULL	
inv_name	text	YES		NULL	
inv_date	date	YES		NULL	
inv_amt	int	YES		NULL	
emp_id	int	YES		NULL	
age	int	YES		NULL	

7. Change the name of the column/field Emp_id to E_id.

```
ALTER TABLE Employee RENAME COLUMN emp_id TO e_id;
DESC Employee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
e_id	int	NO	PRI	NULL	
emp_name	varchar(20)	YES		NULL	
city	varchar(20)	YES		NULL	
bdate	date	YES		NULL	
age	int	YES		NULL	
salary	int	YES		NULL	

8. Update the city of Emp_id-4 to the current city, Nagpur.

```
UPDATE Employee
SET city = 'Nagpur'
WHERE e_id = 104;
SELECT * FROM Employee;
```

Result Grid | Filter Rows: | Edit:

e_id	emp_name	city	bdate	age	salary
101	Mohit	Pune	2005-10-12	NULL	NULL
102	Nikhil	Mumbai	2005-08-08	NULL	NULL
103	Parnika	Nashik	2005-07-30	NULL	NULL
104	Yash	Nagpur	2005-01-26	NULL	NULL
105	Riya	Chennai	2005-08-15	NULL	NULL
NULL	NULL	NULL	NULL	NULL	NULL

9. Delete one record from the Investor table.

```
DELETE
FROM Investor
WHERE inv_no = 2;
```

SELECT * FROM Investor;

Result Grid | Filter Rows: | Edit:

inv_no	inv_name	inv_date	inv_amt	emp_id	age
1	Vikas	2026-08-24	1000	101	NULL
3	Rahul	2024-02-09	78000	102	NULL
4	Virat	2025-09-30	500	103	NULL
5	Shardul	2026-04-21	34000	104	NULL
NULL	NULL	NULL	NULL	NULL	NULL

10. Drop the column 'Age' from the Investor table.

    ALTER TABLE Investor DROP COLUMN age;

    DESC Investor;

Result Grid | Filter Rows: | Export:

Field	Type	Null	Key	Default	Extra
inv_no	int	NO	PRI	NULL	
inv_name	text	YES		NULL	
inv_date	date	YES		NULL	
inv_amt	int	YES		NULL	
emp_id	int	YES		NULL	

11. Drop the CHECK constraint on the 'Age' column from the Employee table.

    ALTER TABLE Employee DROP CONSTRAINT chk_age;

93	14:49:23	ALTER TABLE Employee DROP CONSTRAINT chk_age	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.015 sec

12. Create foreign key constraints between the Employee and Investor tables

    ALTER TABLE Investor ADD FOREIGN KEY (emp_id) REFERENCES Employee(e_id);

Result Grid | Filter Rows: | Export:

Field	Type	Null	Key	Default	Extra
inv_no	int	NO	PRI	NULL	
inv_name	text	YES		NULL	
inv_date	date	YES		NULL	
inv_amt	int	YES		NULL	
emp_id	int	YES	MUL	NULL	

13. Display any 4 columns from the Investor table.

    SELECT inv_no,inv_name,inv_date,inv_amt

    FROM Investor;

Result Grid | Filter Rows:

inv_no	inv_name	inv_date	inv_amt
1	Vikas	2026-08-24	1000
3	Rahul	2024-02-09	78000
4	Virat	2025-09-30	500
5	Shardul	2026-04-21	34000
NULL	NULL	NULL	NULL

14. Rename a table.

    RENAME TABLE Employee TO EmployeeRename;

    DESC EmployeeRename;

Result Grid | Filter Rows: | Export: | Wrap C

Field	Type	Null	Key	Default	Extra
e_id	int	NO	PRI	NULL	
emp_name	varchar(20)	YES		NULL	
city	varchar(20)	YES		NULL	
bdate	date	YES		NULL	
age	int	YES		NULL	
salary	int	YES		NULL	

15. Delete all the records using the TRUNCATE command.

    ALTER TABLE Investor DROP CONSTRAINT investor_ibfk_1;

    TRUNCATE TABLE EmployeeRename;

    TRUNCATE TABLE Investor;

Result Grid | Filter Rows: | Edit:

e_id	emp_name	city	bdate	age	salary
NULL	NULL	NULL	NULL	NULL	NULL

Result Grid | Filter Rows: | Edit:

inv_no	inv_name	inv_date	inv_amt	emp_id
NULL	NULL	NULL	NULL	NULL

16. Drop both tables from the database.

```
DROP TABLE EmployeeRename;
DROP TABLE Investor;
SHOW TABLES;
```

